Abid Ali Karuvally Pathikkal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
GMHSS Perinthalmanna	Class XII (Kerala HSE)	2017	94.75%
MSPEMHS	Class X (Kerala BPE)	2015	10.00%

WORK EXPERIENCE

***Software Engineer* - Commotion**

- **Owned end-to-end development** of an enterprise project from architecture to production; maintained 8 core systems.
- Architected GraphQL data layer with Apollo Client, **cutting API latency** by **30%** via type-safe queries.
- Streamlined application state using Redux Toolkit, **reducing** unnecessary UI **re-renders** by **30%**.
- **Optimized** build pipeline via Vite and TypeScript, **cutting bundle size** by **25%** and speeding reviews.
- Engineered dynamic form engine with Formik and Yup, reducing client-side input errors.

PROJECTS

AI-Guided Computer Vision Physiotherapy Platform

- Engineering real-time CV pipeline tracking 30+ keypoints to deliver instant biomechanical form correction.
- Designing analytics module aggregating range-of-motion (ROM) metrics and patient adherence trends.
- Building end-to-end telemetry system generating structured recovery reports for physical therapists.
- Architecting B2B SaaS platform enabling remote patient monitoring for partner clinics.

Customer Sentiment Analysis of Unacademy

- Scraped and preprocessed 80,000+ app reviews using Python and NLTK for sentiment analytics.
- Engineered LDA topic model and VADER sentiment pipeline to classify core user feedback.
- Quantified user friction points, identifying app latency and subscription pricing as primary churn drivers.

XandY's Strategic Positioning in a Post-Unicorn Era

- Benchmarked K-12 EdTech players across regional and national scale to evaluate unit economics.
- Formulated growth roadmap proposing hyperlocal hybrid micro-centres for middle-school STEM curricula.
- Designed GTM pricing model and transparent brand strategy to capture middle-class market share.

Automatic Tweet Summarization using Transformers

- Architected seq2seq Transformer pipeline using T5 and BART for event-based text summarization.
- Designed custom tokenizer workflow systematically filtering URLs and emojis while retaining key semantics.
- Benchmarked models via ROUGE, achieving top 0.710 ROUGE-1 Recall using Top-p/Top-k sampling.

SKILLS

Programming: Python, SQL, Data Structures, Algorithms, Git

AI/ML: Supervised and Unsupervised Learning, Feature Engineering, Data Preprocessing, Statistical Analysis, Mathematical Modelling, Data Visualization, Deep Learning, LLMs, NLP, Computer Vision

Tools/Framework: Colab, Pytorch, NumPy, Pandas, Scikit-learn, Matplotlib, Tableau

Soft: Communication, Problem Solving, Adaptability, Collaboration, Time Management

RELEVANT & CERTIFICATION COURSES

Mtech: Entrepreneurial **Marketing**, **Innovation** Management, **Game Theory** for Startups, **Operational** Entrepreneurship, **Strategic** Management, **HR** and **Leadership**, Foundation of **Design Thinking** for Entrepreneurship, **Supply Chain** Management and Circular Economy, **Text Analytics** for Business, **Machine Learning** and It's Application,

Btech: **Artificial Intelligence (AI)**, **Computer Vision (CV)**, **Natural Language Processing (NLP)**

Certifications: **Neural Network** Bootcamp (Udemy)

POSITIONS OF RESPONSIBILITY

Hostel Representative | General Council

- Liaised between residents and administration, managing grievance protocols and council operational reporting.
- Resolved daily student escalations with wardens and authorities via structured issue tracking.
- Coordinated hostel logistics and participation for campus-wide events during the Milan festival.

Operations Coordinator | EBSB

- Directed event logistics and vendor operations across 5+ flagship cultural celebrations.
- Led cross-functional execution teams, managing crowd flows and operational safety during major events.
- Optimized campus outreach and resource management to maximize student festival participation.

Logistics & Operations Head | MSME Tech Connect

- Directed end-to-end infrastructure setup and floor allocation for 100+ stalls and 250+ delegates.
- Optimized booth layouts by industry vertical to streamline foot traffic flow across exhibition halls.
- Managed technical asset provisioning, network cabling, and AV infrastructure for all partner booths.

EXTRACURRICULAR ACTIVITIES

Duolingo: Maintaining an unbroken **1,000+ day streak** learning Japanese, French and Chess.

Karate: Awarded **Brown 1st Kyu** and participated in the 11th KBI National Tournament.

Triple Jump: **Silver Medalist** in Perinthalmanna Sub-District Athletics Meet (2016-17).

Tongali Program: Collaborated with a Japanese startup team over a 3-day venture immersion to adapt their product-market strategy for the Indian market, pitching the localized business model to a panel of industry experts and mentors.

Interests & Hobbies: Chess, Coding, Badminton, Football, Travel, Gym